Community Flood Monitoring and Reporting System — Project Proposal

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Project Title

FloodWatch GH: A Community-Driven Flood Reporting System with Severity Prediction

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Problem Statement

Flooding remains one of the most destructive recurring disasters in Ghana. Every rainy season, communities in Accra, Kumasi, Tamale and beyond face property damage, displacement, and loss of life. Yet when a flood occurs, the information available to residents and emergency responders is almost always **incomplete, delayed, and unstructured.**

People rely on WhatsApp forwards and Twitter posts to know which roads are impassable. NADMO responds after damage is already done. And critically, when a flood is reported in one area, **nobody knows how far it is likely to spread** — whether it will stay contained or push into neighboring streets and communities.

That last question is what this project directly answers.

Project Overview

FloodWatch GH is a web-based platform that allows community members to **report flood incidents in real time** — with photos, descriptions, and location — while the system automatically analyzes each report and generates a **flood severity and spread prediction** based on the terrain of the reported area.

The reporting side makes it a community awareness tool. The prediction side makes it an intelligence tool. Together, they create something more useful than either would be alone.

The Core Innovation — Severity Prediction

When a user submits a flood report with a location, the system does the following automatically in the background:

1. **Extracts the coordinates** of the reported incident
2. **Queries elevation data** for that location and its surrounding area using NASA's SRTM dataset (free and publicly available)

3. **Analyzes the terrain gradient** — how steeply or gradually the land slopes away from the reported point
4. **Estimates a spread zone** — the likely area the floodwater could reach if the incident is not contained, based on how water naturally flows downhill across that terrain
5. **Assigns a severity score** — Low, Moderate, High, or Critical — and overlays the estimated spread zone on a map

This turns every community report into a **predictive data point**, not just a record.

How It Works — User Journey

For a resident: A person in Adenta notices their street is flooding. They open FloodWatch GH, drop a pin on their location, write a brief description, optionally attach a photo, and submit. Within seconds, the map updates with their report and shows an estimated spread zone around their area.

For a neighbor or commuter: Someone planning to travel through that area sees the report on the public dashboard, checks the severity score, and decides to take an alternate route.

For a local assembly or NADMO officer: They log into the dashboard and see a live map of active flood reports across their jurisdiction, color-coded by severity, with spread estimates helping them prioritize which areas need immediate attention.

Technical Stack

Component	Technology
Frontend	React / Next.js
Backend	Node.js REST API
Database	PostgreSQL with PostGIS
Mapping	Leaflet.js
Elevation Data	SRTM via OpenTopography API
Spread Estimation	Python script using NumPy and rasterio

Component	Technology
File Storage	Cloudinary (for photo uploads)
Hosting	Vercel + Railway

The Prediction Algorithm — Simplified

The severity prediction does not require a complex machine learning model. It uses a straightforward **flow accumulation approach**:

- Elevation data for the reported area is fetched as a raster grid
- A basic **D8 flow direction algorithm** determines which direction water moves across each cell in the grid
- Cells that receive flow from the reported point are marked as the **predicted spread zone**
- The size and density of that zone determines the severity score

This is academically grounded in hydrology and GIS literature, achievable with open-source Python libraries, and explainable to any examiner without requiring deep machine learning knowledge.

Scope and Deliverables

The project will deliver the following within the academic timeline:

1. A functional web application with user registration and flood incident reporting
2. Photo upload and geotagged location submission per report
3. Automatic elevation data fetching per reported location
4. A working severity scoring and spread estimation module
5. An interactive public map dashboard showing live reports and predicted spread zones
6. A pilot demonstration using **simulated and real reports** across five communities in Greater Accra

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What Makes This Different

Feature	Basic Reporting Tool	FloodWatch GH
Community reporting	✅	✅
Photo uploads	✅	✅
Live map	✅	✅
Severity prediction	❌	✅
Spread zone estimation	❌	✅
Terrain-aware intelligence	❌	✅

The addition of terrain-based severity prediction is the single feature that separates FloodWatch GH from a straightforward CRUD application and positions it as a **decision-support tool** with real-world utility.

Expected Impact

- Residents make faster, informed decisions about evacuation or alternate routes
- Emergency responders gain a prioritized view of where intervention is most urgent
- Local assemblies build a growing database of flood incident history mapped to terrain
- The severity model creates a foundation that can be improved with real historical flood data over time